OCR for Medieval Manuscripts using Transformer-Based Models

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1 Introduction

Optical Character Recognition (OCR) for medieval manuscripts is a challenging problem due to irregular handwriting styles, degraded document quality, and non-standard character representations. Unlike modern printed text, historical manuscripts often include faded ink, noise, and complex layouts, making accurate text extraction difficult.

This project aims to build an OCR system capable of recognizing text from medieval manuscripts using deep learning techniques. A transformer-based approach is adopted, leveraging the TrOCR architecture along with tailored preprocessing steps to handle the unique challenges of historical data.

The final model achieved a Kaggle **public score of 0.55085** and a **private score of 0.57420**, indicating reasonable generalization.

2 Methodology

The overall pipeline consists of:

1. Extracting line-level text regions using XML annotations
2. Applying preprocessing and transformations
3. Training a transformer-based OCR model (TrOCR)

4. Evaluating using Word Error Rate (WER)
5. Generating predictions for test data

3 Data Preprocessing and Transformations

3.1 Line Extraction

Text line regions were extracted from manuscript images using XML annotations. Polygon-based cropping was used where possible, with bounding boxes as fallback. Padding was added to preserve context.

3.2 Image Transformations

The following preprocessing steps were applied:

- Grayscale conversion to simplify representation
- Gaussian blurring to reduce noise
- Adaptive thresholding to enhance contrast
- Dynamic padding to prevent clipping

3.3 Motivation

These transformations improve text visibility, reduce noise, and standardize inputs, making it easier for the model to learn meaningful patterns.

4 Model Architecture

The project uses the TrOCR (Transformer-based OCR) model, which follows an encoder-decoder architecture:

- Encoder: Processes input images

- Decoder: Generates text sequences

Key advantages include handling variable-length sequences, eliminating character segmentation, and leveraging pretrained weights. In particular, we fine-tuned the **trocr-medieval-base pretrained model**, which provides a strong initialization tailored for historical manuscript data.

5 Training Setup

- Epochs: 20
- Optimizer: Adafactor
- Learning Rate: 5e-5
- Batch Size: 64 (train), 16 (validation)
- Evaluation Metric: Word Error Rate (WER)

5.1 Validation Results (Final Epoch)

- Eval Loss: 0.5704
- WER: 0.3258
- Runtime: 768.6 seconds

6 Evaluation Metric

Word Error Rate (WER) is defined as:

$$WER = \frac{S + D + I}{N}$$

where S = substitutions, D = deletions, I = insertions, and N = total words.

7 Results

- Kaggle Public Score: 0.55085
- Kaggle Private Score: 0.57420

The slightly higher private score suggests good generalization.

8 Alternative Models Explored

Several OCR models were evaluated before selecting the final TrOCR-based approach, including PARSeq, different TrOCR variants, PaddleOCR models, the pylaia_catmus_medieval model, and Kraken. Some of these models were character based while the others were word based.

These models did not perform adequately on the medieval manuscript dataset. PARSeq struggled with variability and noise in handwritten text. PaddleOCR models are optimized for modern documents and were unable to handle degraded historical text effectively. The PyLaia medieval model did not generalize well because it was trained on a fixed set of symbols; unknown symbols were mapped to blank tokens due to limitations in the output layer. During finetuning, the model was skipping the cropped images containing unknown symbols for the loss computation. Supporting new symbols would require modifying the network architecture. The Kraken model also did not yield satisfactory results. Some TrOCR variants showed overfitting or unstable validation performance.

These findings reinforced the choice of the final TrOCR configuration.

9 Discussion

9.1 Strengths

- Effective preprocessing for noisy manuscript data
- Strong sequence modeling using transformers
- Robust performance compared to alternative models

10 Future Work

While the current model achieves reasonable performance, several improvements can be explored to further enhance OCR accuracy on medieval manuscripts:

- **Data Augmentation Techniques:** Applying augmentations such as rotation, scaling, noise injection, and contrast variation can help the model generalize better to diverse manuscript styles and degradation patterns.
- **Larger Transformer Models:** Utilizing larger variants of transformer-based architectures may improve sequence modeling and capture more complex patterns in handwritten text.
- **Domain-Specific Fine-Tuning:** Training on additional medieval or historical datasets can help the model adapt better to variations in script, language, and writing styles.
- **Language Model-Based Post-Processing:** Integrating a language model for correcting predicted text can reduce errors by enforcing grammatical and contextual consistency.
- **Vision-Language Models (VLMs):** Exploring modern vision-language models that jointly learn image and text representations could improve performance by incorporating contextual understanding beyond line-level OCR, especially for complex layouts and ambiguous characters.
- **Ensemble Methods:** Combining predictions from multiple models may improve robustness and overall accuracy, especially on difficult samples.

11 Conclusion

This project demonstrates the effectiveness of transformer-based approaches for OCR on medieval manuscripts, a domain characterized by noisy data and highly variable handwriting styles. By combining targeted preprocessing techniques with a TrOCR model, the system is able to achieve competitive performance despite the inherent challenges of the dataset.

The results highlight the importance of both data preparation and model selection in historical OCR tasks. Although the current model performs reasonably well, there remains significant scope for improvement through better data augmentation, model scaling, and post-processing strategies. Overall, this work provides a strong foundation for future exploration in applying deep learning to historical document analysis.